



Executive Summary

Atmospheric Boundary Layer Signal Isolation & Topological Stability

The **SBLToolkit.jl** suite processes complex observational records from nocturnal field tower networks. By leveraging Complex Empirical Orthogonal Functions (CEOF), non-uniform Taylor-Vandermonde derivative operators, and Geometric Surface Pattern Theory (GSPT), the framework cleanly differentiates standing Low-Level Jets (LLJs) from propagating Internal Gravity Waves (IGWs) while mapping saddle-node turbulence collapse manifolds.

CEOF
ACCURACY

99.93%

Wave Wavenumber
Extraction

JET VARIANCE
SHARE

91.37%

Standing Mode 1
Isolation

WAVE VARIANCE
SHARE

8.62%

Propagating Mode 2
Isolation

FOLD SHEAR
 S_{FOLD}

0.5898

Analytical
Bifurcation (s^{-1})

FOLD TKE E_{FOLD}

0.2156

Critical Energy ($\text{m}^2 \text{s}^{-2}$)

1. Field Campaign Comparative Analysis

Evaluating tower spatial resolution, vertical level density (N_z), and GSPT inversion feasibility.

GSPT transition inversions depend on high vertical tower resolution to construct non-uniform spatial derivative matrices ($D1$, $D2$). While multi-level masts like CASES-99 (55m, $N_z=7$) and GABLS3 (200m, $N_z=20$) yield full topological stability mappings, sparse arrays like SHEBA (2 levels) trigger a *Bypassed Guard* due to stencil under-determination.

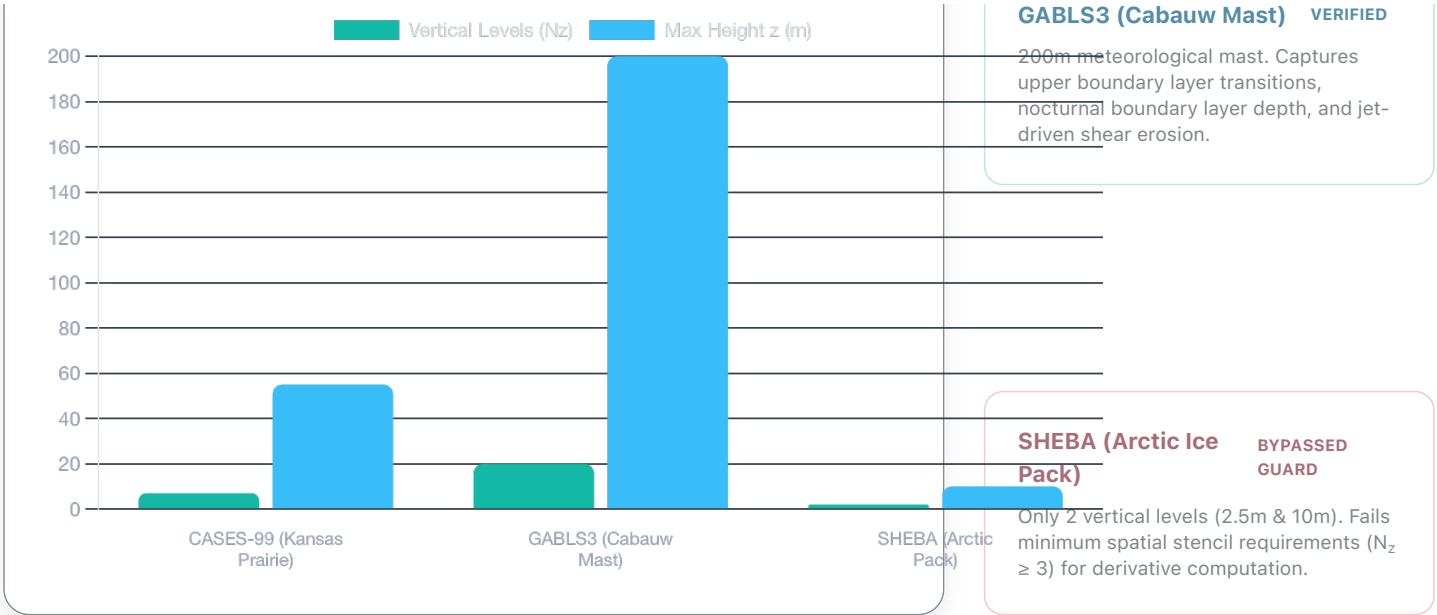


Tower Height & Vertical Grid Density Comparison

CASES-99 (Kansas Prairie)

VERIFIED

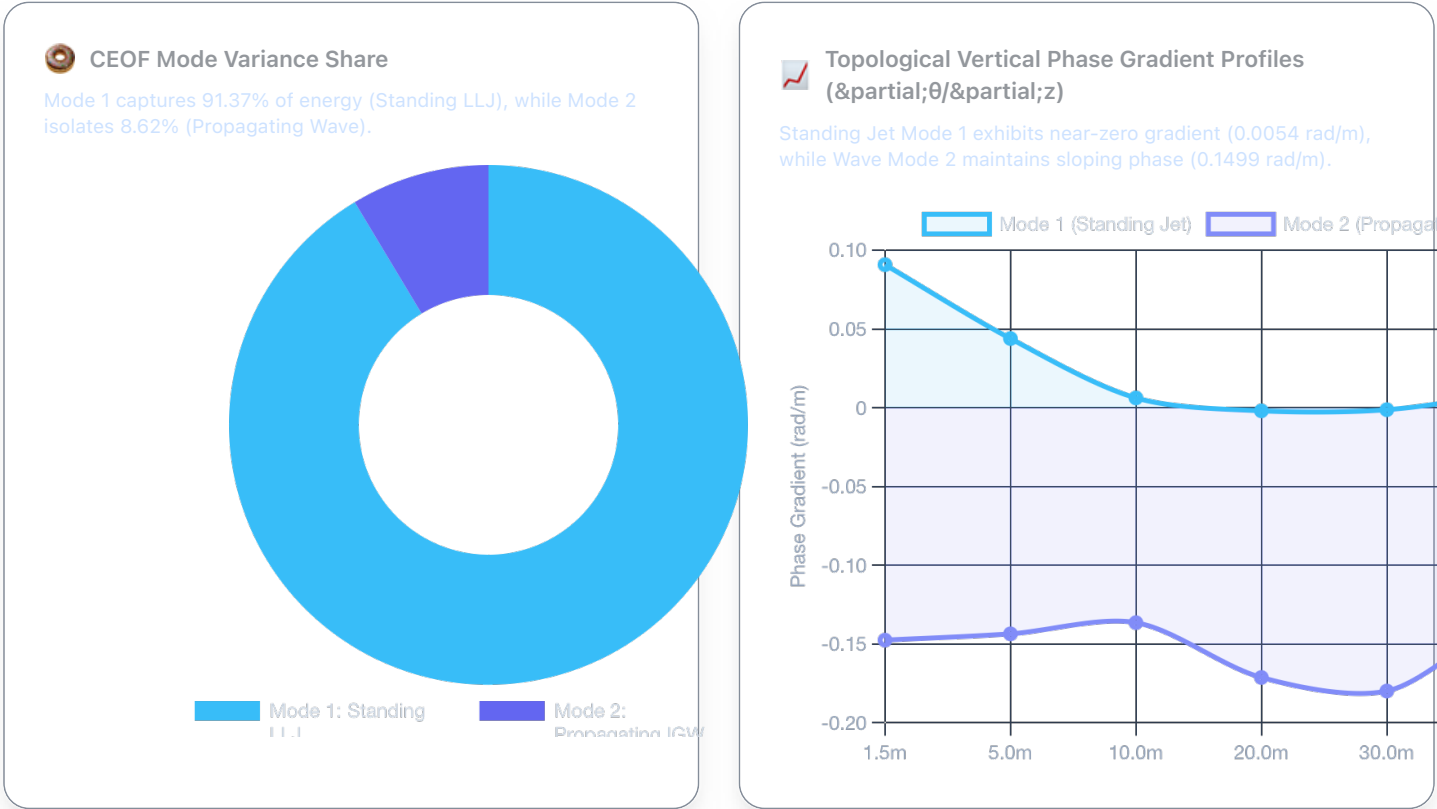
55m tower with 7 non-uniform heights (1.5m to 55m). Ideal for capturing nocturnal Low-Level Jet nose formation near 30m.



2. Complex EOF Wave-Jet Separation Pipeline

Decomposing velocity anomaly matrices into standing jet modes and propagating internal gravity waves.

Using a self-contained matrix Fourier Hilbert transform, real velocity fields are converted to complex analytic signals $z(t) = x(t) + i H[x(t)]$. Complex SVD extracts spatial mode amplitudes and vertical phase profiles. Low-amplitude phase noise near the surface is neutralized using an **amplitude-weighted vertical phase gradient metric**.



3. Interactive GSPT Saddle-Node Simulator & Fast-Slow Relaxation Engine



SBL Toolkit & GSPT Framework

Stable Boundary Layer Signal Separation & Topological Bifurcation

[Overview](#)[Field Campaigns](#)[CEOF Wave-Jet](#)[Saddle-Node Dynamics](#)[Regularization](#)[QC Pipeline](#)

Real-Time Numerical Parameter Controls

Adjust geostrophic forcing, relaxation times, and mixing length to recompute trajectory paths and critical fold bounds.

S_{fold} : **0.58985**
 s^{-1}

e_{fold} : **0.21556**
 m^2s^{-2}

Geostrophic Forcing (G_0):

0.30

Relaxation Parameter (ϵ):

0.05

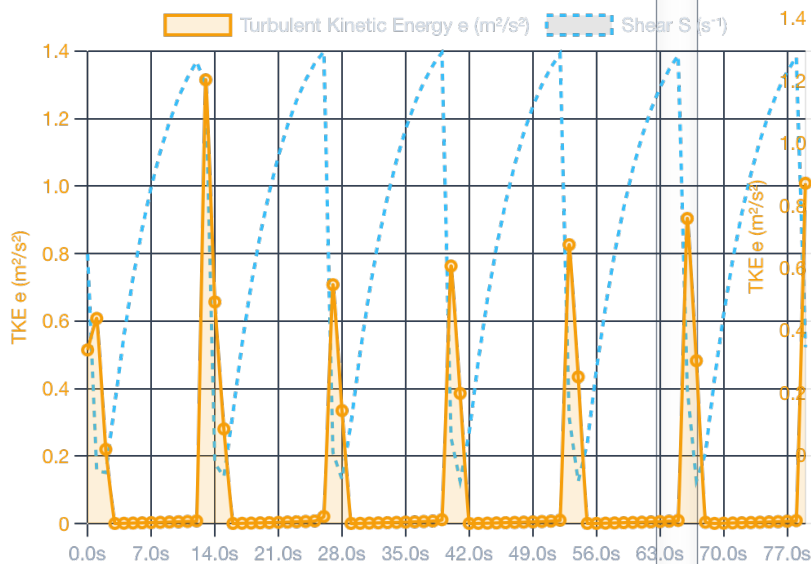
Mixing Length (l_0):

1.00



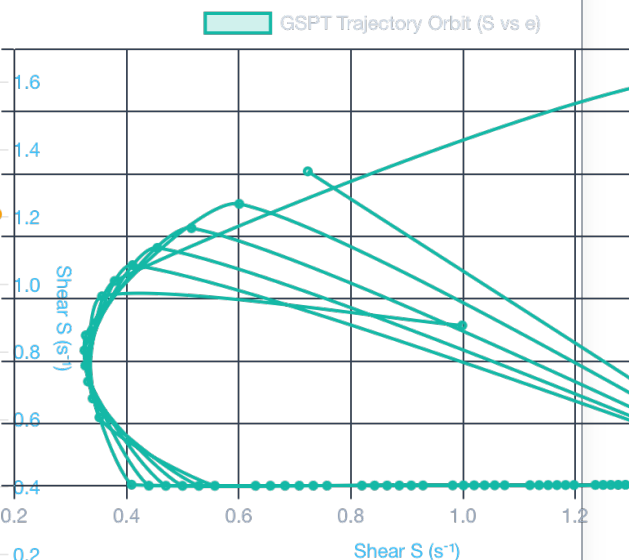
Time Series Trajectory: Shear (S) & TKE (e) Relaxation Cycle

Forward time integration showing sudden TKE collapses and slow wind shear rebuild phases.



2D Phase Portrait: Shear (S) vs. TKE (e) Orbit

Phase-space trajectory cycling around the critical fold line S_{fold} into the laminar regime.



ANALYTICAL SADDLE-NODE FOLD

- Critical Fold Shear: **0.58985** s^{-1}
- Critical Fold TKE: **0.21556** m^2s^{-2}
- Physical Mechanism: Saddle-Node manifold annihilation under strong radiative cooling.

HYSTERETIC STATE PARTITION

- Stable Turbulent State:** TKE $e > 0.20$ m^2s^{-2} (Active shear mixing)
- Laminar Attractor:** TKE $e < 0.01$ m^2s^{-2} (Turbulence collapse state)

JET SHEAR REVIVAL CYCLE

During laminar collapse, friction drops, allowing the Low-Level Jet to accelerate. Shear S accumulates until exceeding the upper ignition threshold, triggering intermittent turbulent bursts.

4. Numerical Stability & Singular Value Mitigation

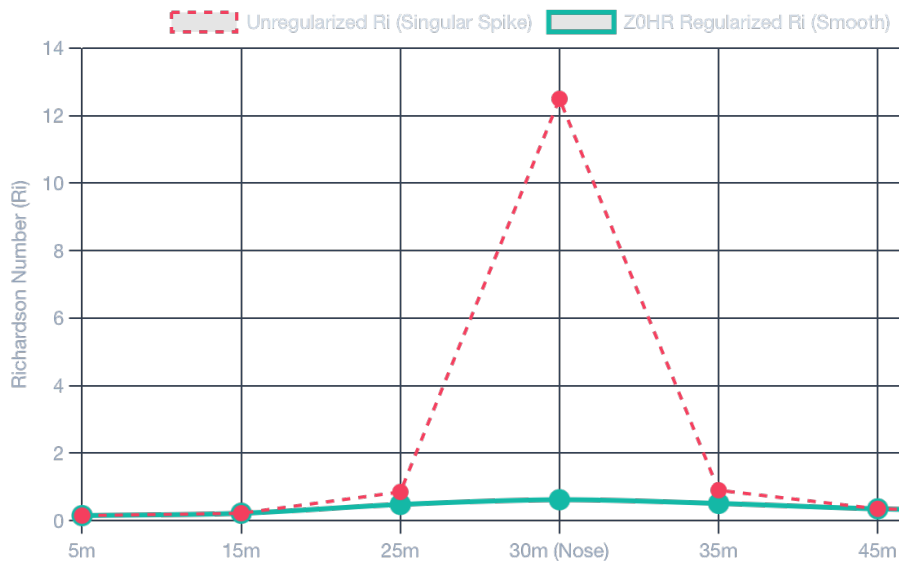
Preventing runaway cooling bifurcations and division-by-zero quotient singularities near Jet cores.

Under strong nocturnal cooling, the gradient Richardson number $Ri = (g/\theta_v \cdot \partial \theta_v / \partial z) / [(\partial u / \partial z)^2 + (\partial v / \partial z)^2]$ suffers from division-by-zero singularities near the LLJ nose where wind shear vanishes ($\partial u / \partial z \rightarrow 0$). SBLToolkit.jl applies Tikhonov-Morozov filtering and Zero-Offset Hyperbolic Regularization (Z0HR) to guarantee bounded numerical updates.



Richardson Number Profile: Raw vs Z0HR Regularized

Unregularized Ri spikes to infinity at the Jet nose (30m), whereas Z0HR maintains smooth, differentiable stability bounds.



TRACK A PRIMITIVE REGULARIZATION

Applies Tikhonov-Morozov filtering directly to velocity and potential temperature profiles under a Modified Morozov Discrepancy Principle (MDP). Matches smoothing scale to noise floor δ .

ZERO-OFFSET HYPERBOLIC SCHEME (Z0HR)

Replaces singular quotient denominators with hyperbolic smoothing operators $\sqrt{x^2 + \epsilon^2}$. Eliminates division-by-zero without altering physical equilibria.

TANGENTIAL CONE CONDITION

Ensures non-linear stability bound $\tau_2 \geq \tau_1(3 + 2\gamma)$, guaranteeing a unique regularization parameter α for non-linear inverse convergence.

5. Quality Control Gating & GSPT Framework Architecture

Triple-Point Dispersion QC validation and multi-phase prognostic GSPT integration.

To prevent silent numerical corruption from creating false regime transitions, SBLToolkit.jl enforces a strict Quality Control pipeline using Triple-Point Dispersion (δ_{TP}) alongside schema adapters (``src/ultra/adapters/``) to sanitize incoming NetCDF tower streams.

1 Diffusivity Cutoff Height

Vertical height where turbulent eddy diffusivity K_m drops below physical mixing threshold ($10^{-4} \text{ m}^2/\text{s}$).

Marker $z_K : K_m(z) \rightarrow \epsilon$

2 TKE Level-Set Height

Height level where Turbulent Kinetic Energy drops below critical threshold (e.g. $e = 0.05 \text{ m}^2/\text{s}^2$).

Marker $z_e : TKE(z) = e_{crit}$

3 TKE Gradient Extremum

Point of maximum TKE vertical collapse gradient ($\partial e / \partial z = \max$), marking active collapse front.

Marker $z_{grad} : \max(|\partial TKE / \partial z|)$



GSPT Evolutionary Phases: Diagnostic Mapping to Prognostic Closure

PHASE 1

Data Cleansing & Regularization

Planar fit rotations, u- flatline replacements (0.3 m/s baseline), and Track A primitive regularization.

PHASE 2

Diagnostic Inversion (R_{coord})

Maps profiles onto topological manifold coordinate R_{coord} to distinguish shear mixing from laminar collapse.

PHASE 3

Prognostic Closure Integration

Embeds R_{coord} manifold geometries into 1.5-order & 2nd-order turbulence models for jet shear revival prediction.